

HPC

Why is the program used?

It is used to process large datasets faster by dividing computations among multiple CPU threads, improving performance and reducing execution time.

Corrected: This whole code should come in a single line

```
#pragma omp parallel for reduction(min:min_val) reduction(max:max_val)  
reduction(+:sum)
```

1. What is Parallel Computing?

Parallel Computing is the execution of multiple tasks simultaneously using multiple processors or cores to reduce execution time.

2. What is HPC?

High Performance Computing (HPC) uses powerful computers and parallel processing techniques to solve large and complex computational problems faster.

3. What is Parallel Reduction?

Parallel Reduction is a technique where multiple threads process data simultaneously to produce a single result such as sum, minimum, maximum, or average.

4. What mathematical properties must a reduction operator possess?

A reduction operator should satisfy:

- Associative Property
- Commutative Property

These properties ensure correct results during parallel execution.

5. Difference between Sequential and Parallel Reduction?

Sequential Reduction

Processes one element at a time

Uses single CPU core

Slower execution

Lower performance for large data

Parallel Reduction

Processes multiple elements simultaneously

Uses multiple CPU cores

Faster execution

Better performance for large data

Sequential Reduction

Time Complexity: $O(n)$

Parallel Reduction

Time Complexity: $O(\log n)$

6. What is a Race Condition?

A Race Condition occurs when multiple threads access and modify the same variable simultaneously, causing incorrect or unpredictable results.

7. How does Parallel Reduction avoid Race Condition?

Parallel Reduction uses private thread variables and OpenMP reduction() to safely combine results without conflicts.

8. What is OpenMP?

OpenMP is an API used for shared-memory parallel programming in C/C++ using multithreading.

9. What is CUDA?

CUDA stands for **Compute Unified Device Architecture**.

CUDA is a parallel computing platform and GPU programming model developed by NVIDIA used for high-speed parallel processing and GPU computations.

10. What is a Thread?

A Thread is the smallest unit of execution within a process.

11. Types of Parallelism?

- Data Parallelism
- Task Parallelism

12. Difference between CPU and GPU?

CPU – Central Processing Unit

Performs fewer complex tasks

Has fewer powerful cores

Better for sequential processing

Used for general computing

GPU – Graphics Processing Unit

Performs many parallel tasks simultaneously

Has thousands of smaller cores

Better for parallel processing

Used for AI, graphics, and HPC

Which is better?

- CPU is better for logical and general-purpose tasks.

- GPU is better for parallel computations and large data processing.

13.Simple Applications of Parallel Reduction?

- Adding marks of students
- Finding highest salary in a company
- Calculating total likes or views on social media

14.Technical Applications of Parallel Reduction?

- Image processing
- AI and Machine Learning computations
- Weather and climate analysis

15.Applications of HPC?

HPC is used in weather forecasting, scientific simulations, AI training, cryptography, image processing, and big data analysis to solve complex problems faster.

DL

1. What is Artificial Intelligence (AI)?

AI is the ability of machines to perform tasks that normally require human intelligence.

2. What is Machine Learning?

Machine Learning is a subset of AI where systems learn patterns automatically from data.

3. What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses multilayer neural networks for learning complex patterns.

4. What is a Neural Network?

A Neural Network is a system of interconnected artificial neurons used for learning and prediction.

5. What is DNN?

Deep Neural Network (DNN) is a neural network containing multiple hidden layers between input and output layers.

6. How does DNN work in classification?

DNN learns patterns from training data and predicts the correct category or class label for input data.

7. What is Classification?

Classification is the process of categorizing data into predefined classes or groups.

8. Example of Classification?

Spam email detection, face recognition, and disease prediction.

9. What is Binary Classification?

Binary Classification divides data into two categories such as Yes/No or Spam/Not Spam.

10. What is Binary Cross Entropy?

Binary Cross Entropy is a loss function used to measure error in binary classification models.

11. What is an Epoch?

An Epoch is one complete training cycle in which the entire dataset passes through the neural network once.

12. What is the Epoch Cycle?

The Epoch Cycle refers to repeatedly training the model on the dataset to improve accuracy.

13. What is Validation Split?

Validation Split separates a portion of training data for validation during model training.

14. What is Adam Optimizer?

Adam Optimizer is an optimization algorithm used to improve model training speed and accuracy.

15. What is CNN?

Convolutional Neural Network (CNN) is mainly used for image processing and computer vision tasks.

16. What is RNN?

Recurrent Neural Network (RNN) is used for sequential data such as text, speech, and time-series data.

17. What is TensorFlow?

[TensorFlow](#) is an open-source Machine Learning and Deep Learning framework developed by Google for building and training neural network models.

18. What is Keras?

[Keras](#) is a high-level Deep Learning API built on TensorFlow used for easily creating and training neural networks using simpler code.

19. What is Overfitting?

Overfitting occurs when a model performs very well on training data but poorly on new unseen data.

20. What is Underfitting?

Underfitting occurs when a model is too simple to learn training data properly, resulting in poor accuracy.

21. What is NumPy and its use?

[NumPy](#) is a Python library used for numerical operations and array processing.

22. What is Pandas and its use?

[Pandas](#) is a Python library used for data analysis and handling datasets.

23. What is Matplotlib and its use?

[Matplotlib](#) is a Python library used for plotting graphs and data visualization.

24. What is Seaborn and its use?

[Seaborn](#) is a Python visualization library used for creating statistical and attractive graphs.